

RESERCH METHODOLOGY

MODULE ONE

Meaning of research

According to Perves, Ghauri, Kjell, Gronhang (2002), **research is a process of planning, executing and investigating in order to find out specific questions.**

In order to get reliable answers to our questions we need to do this kind of investigation in a systematic manner, so that it is easier for others to understand the logic of and believe in our report.

Likewise, Kothari (2004) argues that research in common parlance refers **to a search for knowledge**. Or can be define research as a **scientific and systematic search for pertinent information on a specific topic**. In fact, research is **an art of scientific investigation**.

The Advanced Learner's Dictionary of Current English lays down the meaning of research as **“a careful investigation or inquiry specially through search for new facts in any branch of knowledge.”**

This helps Redman and Mory to define research as a **“systematized effort to gain new knowledge.”** This has different views to other people who consider research as **a movement, a movement from the known to the unknown**. It is actually a voyage of discovery. We all possess the vital instinct of inquisitiveness for, when the unknown confronts us, we wonder and our inquisitiveness makes us probe and attain full and fuller understanding of the unknown.

Generally, **Research is a systematic and scientific inquiry or investigation aimed at discovering, interpreting, and controlling observed phenomena**. It involves collecting data, documenting critical information, and analyzing and

interpreting that data/information using suitable methodologies set by specific professional fields and academic disciplines.

OBJECTIVES OF THE RESEARCH

Research objectives describe what your research project intends to accomplish. They should guide every step of the research process, including how you collect data, build your argument, and develop your conclusions.

- i. To gain familiarity with a phenomenon or to achieve new insights into it.
- ii. To portray accurately the characteristics of a particular individual, situation or a group.
- iii. To determine the frequency with which something occurs or with which it is associated with something else.

Research Ethics and Implications

Ethical considerations in research are **a set of principles that guide your research designs and practices**. Scientists and researchers must always **adhere to a certain code of conduct** when collecting data from people. The goals of human research often include understanding real-life phenomena, studying effective treatments, investigating behaviours, and improving lives in other ways. What you decide to research and how you conduct that research involve key ethical considerations. These considerations work to protect **the rights of research participants, enhance research validity, maintain scientific or academic integrity**.

Types of ethical issues

There are several ethical issues you should always pay attention to in your research design, and these issues can overlap with each other. You'll usually outline ways you'll deal with each issue in your research proposal if you plan to collect data from participants.

i) Voluntary participation

Voluntary participation means that all research subjects are free to choose to participate without any pressure or coercion.

All participants are able to withdraw from, or leave, the study at any point without feeling an obligation to continue. Your participants don't need to provide a reason for leaving the study.

It's important to make it clear to participants that there are no negative consequences or repercussions to their refusal to participate. After all, they're taking the time to help you in the research process, so you should respect their decisions without trying to change their minds. Voluntary participation is an ethical principle protected by international law and many scientific codes of conduct. Take special care to ensure there's no pressure on participants when you're working with vulnerable groups of people who may find it hard to stop the study even when they want to.

Example of voluntary participation "When recruiting participants for an experiment, you inform all potential participants that they are free to choose whether they want to participate, and they can withdraw from the study anytime without any negative repercussions".

ii) Informed consent

Informed consent refers to a situation in which all potential participants receive and understand all the information they need to decide whether they want to participate. This includes information about the study's benefits, risks, funding, and institutional approval.

Example of informed consent "You recruit participants outside a train station for a quick survey.

You make sure to provide all potential participants with all the relevant information about

- ❖ What the study is about
- ❖ The risks and benefits of taking part
- ❖ How long the study will take
- ❖ Your supervisor's contact information and the institution's approval number"

You also let them know that their data will be kept confidential, and they are free to stop filling in the survey at any point for any reason. They can also withdraw their information by contacting you or your supervisor.

Usually, you'll provide participants with a text for them to read and ask them if they have any questions. If they agree to participate, they can sign or initial the consent form. Note that this may not be sufficient for informed consent when you work with particularly vulnerable groups of people.

If you're collecting data from people with low literacy, make sure to verbally explain the consent form to them before they agree to participate.

For participants with very limited English proficiency, you should always translate the study materials or work with an interpreter so they have all the information in their first language.

In research with children, you'll often need informed permission for their participation from their parents or guardians. Although children cannot give informed consent, it's best to also ask for their assent (agreement) to participate, depending on their age and maturity level.

iii) Anonymity

Anonymity means that you don't know who the participants are and you can't link any individual participant to their data.

You can only guarantee anonymity by not collecting any personally identifying information for example, names, phone numbers, email addresses, IP addresses, physical characteristics, photos, and videos.

In many cases, it may be impossible to truly anonymize data collection. For example, data collected in person or by phone cannot be considered fully anonymous because some personal identifiers (demographic information or phone numbers) are impossible to hide.

You'll also need to collect some identifying information if you give your participants the option to withdraw their data at a later stage.

Data pseudonymization is an alternative method where you replace identifying information about participants with pseudonymous, or fake, identifiers. The data can still be linked to participants but it's harder to do so because you separate personal information from the study data.

Example of data pseudonymization “You’re conducting a survey with college students. You ask participants to enter demographic information including their age, gender identity, nationality, and ethnicity. With all this information, it may be possible for other people to identify individual participants, so you pseudonymize the data. Each participant is given a random three-digit number. You separate their personally identifying information from their survey data and include the participant numbers in both files. The survey data can only be linked to personally identifying data via the participant numbers”.

iv) Confidentiality

Confidentiality means that you know who the participants are, but you remove all identifying information from your report. All participants have a right to privacy, so you should protect their personal data for as long as you store or use it. Even when you can't collect data anonymously, you should secure confidentiality whenever you can.

Example of confidentiality “To keep your data confidential, you take steps to safeguard it and prevent any threats to data privacy. You store all signed consent forms in a locked file drawer, and your password-protect all files with survey data”.

Only other researchers approved by the IRB (institutional review board) are allowed to access the study data, and you make sure that everyone knows and follows your institution's data privacy protocols. Some research designs aren't conducive to confidentiality, but it's important to make all attempts and inform participants of the risks involved.

Example of focus group confidentiality “In a focus group study, you invite five people to give their opinions on a new student service in a group setting”.

Before beginning the study, you ask everyone to agree to keep what's discussed confidential and to respect each other's privacy. You also note that you cannot completely guarantee confidentiality or anonymity so that participants are aware of the risks involved.

v) Potential for Harm

As a researcher, you have to consider all possible sources of harm to participants. Harm can come in many different forms.

- **Psychological harm:** Sensitive questions or tasks may trigger negative emotions such as shame or anxiety.
- **Social harm:** Participation can involve social risks, public embarrassment, or stigma.
- **Physical harm:** Pain or injury can result from the study procedures.
- **Legal harm:** Reporting sensitive data could lead to legal risks or a breach of privacy.

It's best to consider every possible source of harm in your study as well as concrete ways to mitigate them. Involve your supervisor to discuss steps for harm reduction. Make sure to disclose all possible risks of harm to participants before the study to get informed consent. If there is a risk of harm, prepare to provide participants with resources or counselling or medical services if needed.

Example of potential for harm “In a study on stress, you survey college students on their alcohol consumption habits.

Some of these questions may bring up negative emotions, so you inform participants about the sensitive nature of the survey and assure them that their responses will be confidential”.

You also provide participants with information about student counselling services and information about managing alcohol use after the survey is complete.

vi) Results Communication

The way you communicate your research results can sometimes involve ethical issues. Good science communication is honest, reliable, and credible. It's best to make your results as transparent as possible. **Take steps to actively avoid plagiarism and research misconduct wherever possible.**

a) Plagiarism

Plagiarism means submitting others' works as your own. Although it can be unintentional, copying someone else's work without proper credit amounts to stealing. It's an ethical problem in research communication because you may benefit by harming other researchers.

Self-plagiarism is when you republish or re-submit parts of your own papers or reports without properly citing your original work.

This is problematic because you may benefit from presenting your ideas as new and original even though they've already been published elsewhere in the past. You may also be infringing on your previous publisher's copyright, violating an ethical code, or wasting time and resources by doing so.

In extreme cases of self-plagiarism, entire datasets or papers are sometimes duplicated. These are major ethical violations because they can skew research findings if taken as original data.

Example of duplication “You're conducting a meta-analysis on whether working from home is related to better stress management. You gather all studies on this topic that meet your search criteria.

You notice that two published studies have similar characteristics even though they are from different years. Their sample sizes, locations, treatments, and results are highly similar, and the studies share one author in common.

If you enter both data sets in your analyses, you get a different conclusion compared to when you only use one data set. Including both data sets would distort your overall findings”.

b) Research misconduct

Research misconduct means making up or falsifying data, manipulating data analyses, or misrepresenting results in research reports. It’s a form of academic fraud. These actions are committed intentionally and can have serious consequences; research misconduct is not a simple mistake or a point of disagreement about data analyses.

Research misconduct is a serious ethical issue because it can undermine academic integrity and institutional credibility. It leads to a waste of funding and resources that could have been used for alternative research.

Example of misconduct (MMR vaccine misinformation) In 1998, Andrew Wakefield and others published a now-debunked paper claiming that the measles, mumps, and rubella (MMR) vaccine causes autism in children. Later investigations revealed that they fabricated and manipulated their data to show a non-existent link between vaccines and autism. Wakefield also neglected to disclose important conflicts of interest, and his medical license was taken away. This fraudulent work sparked vaccine hesitancy among parents and caregivers. The rate of MMR vaccinations in children fell sharply, and measles outbreaks became more common due to a lack of herd immunity. In reality, there is no risk of children developing autism from the MMR or other vaccines, as shown by many large studies. Although the paper was retracted, it has actually received thousands of citations”.

EXAMPLES OF ETHICAL FAILURES

Tuskegee syphilis study “The Tuskegee syphilis study was an American public health study that violated research ethics throughout its 40-year run from 1932 to 1972. In this study, 600 young black men were deceived into participating with a promise of free healthcare that was never fulfilled.

In reality, the actual goal was to study the effects of the disease when left untreated, and the researchers never informed participants about their diagnoses or the research aims.

Although participants experienced severe health problems, including blindness and other complications, the researchers only pretended to provide medical care. When treatment became possible in 1943, 11 years after the study began, none of the participants were offered it, despite their health conditions and high risk of death.

By the end of the study, 128 participants had died of syphilis or related complications. The study ended only once its existence was made public and it was judged to be “medically unjustified.”

Ethical failures like these resulted in severe harm to participants, wasted resources, and lower trust in science and scientists. This is why all research institutions have strict ethical guidelines for performing research.

Reference

Bhandari, P. (2023, June 22). *Ethical Considerations in Research | Types & Examples*. Scribbr. Retrieved October 24, 2023, from <https://www.scribbr.com/methodology/research-ethics/>

Ethical issues in the Researcher-Participant relationship

The researcher -participant (subject) relationship became the most sensitive one in the process of research in business studies. It is obvious that researcher struggle much with the decision of whether to inform the participant about the real purpose of the research. They are believing that telling them the whole truth might result in the participant refusing or being reluctant to give full cooperation. Churchill (1999:50) presents ethical concerns on eight areas of the researchers-participant relationship as follow:

1. Preserving participant's anonymity
2. Do not asking participants questions detrimental to their self-interest
3. Use of special equipment and techniques, e.g., tape recorder, video
4. Do not use of coercion to get information
5. Do not depriving participants of their rights, e.g., of self-determination
6. Do not exposing participant to mental stress
7. Do not involving participants in research without their consent
8. Do not use of deception

TYPES OF RESEARCH

Classification of Types of Research

There are various types of research that are classified according to their objective, depth of study, analyzed data, time required to study the phenomenon and other factors. It's important to note that a research project will not be limited to one type of research, but will likely use several.

According to its Purpose

i) Theoretical Research

Theoretical research, also referred to as pure or basic research, focuses on generating knowledge, regardless of its practical application. Here, data collection is used to generate new general concepts for a better understanding of a particular field or to answer a theoretical research question.

Results of this kind are usually oriented towards the formulation of theories and are usually based on documentary analysis, the development of mathematical formulas and the reflection of high-level researchers.

For example, a philosophical dissertation, since the aim is to generate new approaches from existing data without considering how its findings can be applied or implemented in practice.

ii) Applied Research

Here, the goal is to find strategies that can be used to address a specific research problem. Applied research draws on theory to generate practical scientific knowledge, and its use is very common in STEM fields such as engineering, computer science and medicine.

This type of research is subdivided into two types:

1. **Technological applied research:** looks towards improving efficiency in a particular productive sector through the improvement of processes or machinery related to said productive processes.
2. **Scientific applied research:** has predictive purposes. Through this type of research design, we can measure certain variables to predict behaviours useful to the goods and services sector, such as consumption patterns and viability of commercial projects.

For example, market research, because by examining consumption patterns, strategies can be developed for the development of new products and marketing campaigns, etc.

Note: Applied research is usually based on knowledge or results obtained through theoretical research.

In fact, it is common for research projects to first establish the theoretical framework both to define the field of study and to identify possible theories that could be tested or applied to solve the specific problem posed in the project.

According to your Depth of Scope

i) Exploratory Research

Exploratory research is used for the preliminary investigation of a subject that is not yet well understood or sufficiently researched. It serves to establish a frame of reference and a hypothesis from which an in-depth study can be developed that will enable conclusive results to be generated.

Because exploratory research is based on the study of little-studied phenomena, it relies less on theory and more on the collection of data to identify patterns that explain these phenomena.

For example, an investigation of the role social media in the perception of self-image.

ii) Descriptive Research

The primary objective of descriptive research is to define the characteristics of a particular phenomenon without necessarily investigating the causes that produce it.

In this type of research, the researcher must take particular care not to intervene in the observed object or phenomenon, as its behaviour may change if an external factor is involved.

For example, investigating how the public census of influential government officials differs between urban and non-urban areas.

iii) Explanatory Research

Explanatory research is the most common type of research method and is responsible for establishing cause-and-effect relationships that allow generalisations to be extended to similar realities. It is closely related to descriptive research, although it provides additional information about the observed object and its interactions with the environment.

For example, investigating the brittle behaviour of a specific material when under compressive load.

iv) Correlational Research

The purpose of this type of scientific research is to identify the relationship between two or more variables. A correlational study aims to determine whether a variable changes, how much the other elements of the observed system change.

According to the Type of Data Used

i) Qualitative Research

Qualitative methods are often used in the social sciences to collect, compare and interpret information, has a linguistic-semiotic basis and is used in techniques such as discourse analysis, interviews, surveys, records and participant observations.

In order to use statistical methods to validate their results, the observations collected must be evaluated numerically. Qualitative research, however, tends to be subjective, since not all data can be fully controlled. Therefore, this type of research design is better suited to extracting meaning from an event or phenomenon (the ‘why’) than its cause (the ‘how’).

For example, examining the effects of sleep deprivation on mood.

ii) Quantitative Research

Quantitative research study delves into a phenomenon through quantitative data collection and using mathematical, statistical and computer-aided tools to measure them. This allows generalised conclusions to be projected over time.

For example, conducting a computer simulation on vehicle strike impacts to collect quantitative data.

According to the Degree of Manipulation of Variables

i) Experimental Research

It is about designing or replicating a phenomenon whose variables are manipulated under strictly controlled conditions in order to identify or discover its effect on another independent variable or object. The phenomenon to be studied is measured through study and control groups, and according to the guidelines of the scientific method.

For example, randomised controlled trial studies for measuring the effectiveness of new pharmaceutical drugs on human subjects.

ii) Non-Experimental Research

Also known as an observational study, it focuses on the analysis of a phenomenon in its natural context. As such, the researcher does not intervene directly, but limits their involvement to measuring the variables required for the study. Due to its observational nature, it is often used in descriptive research.

For example, a study on the effects of the use of certain chemical substances in a particular population group can be considered a non-experimental study.

iii) Quasi-Experimental Research

It controls only some variables of the phenomenon under investigation and is therefore not entirely experimental. In this case, the study and the focus group cannot be randomly selected, but are chosen from existing groups or populations. This is to ensure the collected data is relevant and that the knowledge, perspectives and opinions of the population can be incorporated into the study.

For example, assessing the effectiveness of an intervention measure in reducing the spread of antibiotic-resistant bacteria.

According to the Type of Inference

i) Deductive Investigation

In this type of research, reality is explained by general laws that point to certain conclusions; conclusions are expected to be part of the premise of the research problem and considered correct if the premise is valid and the inductive method is applied correctly.

ii) Inductive Research

In this type of research, knowledge is generated from an observation to achieve a generalisation. It is based on the collection of specific data to develop new theories.

Hypothetical-Deductive Investigation

It is based on observing reality to make a hypothesis, then use deduction to obtain a conclusion and finally verify or reject it through experience.

According to the Time in which it is Carried Out

i) Longitudinal Study (also referred to as Diachronic Research)

It is the monitoring of the same event, individual or group over a defined period of time. It aims to track changes in a number of variables and see how they evolve over time. It is often used in medical, psychological and social areas.

For example, a cohort study that analyses changes in a particular indigenous population over a period of 15 years.

ii) Cross-Sectional Study (also referred to as Synchronous Research)

Cross-sectional research design is used to observe phenomena, an individual or a group of research subjects at a given time.

According to the Sources of Information

i) Primary Research

This fundamental research type is defined by the fact that the data is collected directly from the source, that is, it consists of primary, first-hand information.

ii) Secondary research

Unlike primary research, secondary research is developed with information from secondary sources, which are generally based on scientific literature and other documents compiled by another researcher.

RESEARCH PROCESS

The research process consists of several interconnected parts that a researcher must follow to generate valuable knowledge and focus on a relevant topic. The main parts of the research process include:

i) Identifying and defining the research problem:

This step involves understanding the research challenge, setting clear goals, and formulating research questions or hypotheses. It is crucial to grasp exactly what you are attempting to learn and what you want to achieve

ii) Literature review:

This stage involves reviewing existing literature and research on the topic to gain a better understanding of the problem, identify research gaps, and build consistency between your work and others. The knowledge gained through the review of literature guides the researcher in clarifying and narrowing the research project

iii) Research design:

In this step, the researcher selects the appropriate research design, which can be experimental, survey research, correlational, semi-experimental, or review, depending on the research problem

The research design process is a systematic and structured approach to conducting research, ensuring that the study is well-planned, ethical, and rigorous

iv) Data collection:

This step involves gathering relevant data using various methods, such as surveys, interviews, observations, or experiments, depending on the research design. The data collected should be valid and reliable, ensuring that it produces meaningful results

v) Data analysis:

Once the data is collected, it needs to be analyzed using appropriate techniques, such as statistical analysis, content analysis, or discourse analysis, to interpret the results and draw conclusions

vi) Interpretation and reporting of findings:

In this step, the researcher interprets the results, compares them with existing knowledge, and draws conclusions. The findings are then reported in a clear and concise manner, often in the form of a research paper, presentation, or report

By following these steps, researchers can ensure that their study is well-structured, ethical, and valuable, contributing to the existing body of knowledge on the topic.

MODULE TWO

MEANING OF DATA

Data refers to information, such as facts, statistics, or numbers that is collected, processed, and used for analysis, decision-making, or to support reasoning and discussion. In the context of computing, data is information that has been translated into a form that is efficient for movement or processing. The term "data" is derived from the Latin word "datum," which means "something given".

Key points about data include:

- ❖ Data can be in various forms, such as text, numbers, images, audio, or video, and can be stored and processed electronically.
- ❖ In scientific and academic writing, data is often treated as a plural noun, while in other types of writing, it can be either singular or plural.
- ❖ The concept of big data refers to a large volume of data that is characterized by its volume, variety, and velocity, and has led to the development of new business models and concerns about data privacy.
- ❖ Data has meaning beyond its use in computing and is also relevant in fields such as finance, marketing, demographics, and health, where it describes a gathered body of facts.

TYPES OF DATA

There are different types of data that are used in researches. The main types of data include:

1. Qualitative or categorical data:

This type of data is non-numerical and is used to describe characteristics or qualities of a subject. It can be further classified into nominal and ordinal data.

- ❖ **Nominal data:** This type of data is used to label variables without providing numerical values. Examples include gender, race, or religion.
- ❖ **Ordinal data:** This type of data is used to rank variables in a specific order. Examples include education level or income bracket.

2. Quantitative or numerical data:

This type of data is numerical and can be measured. It can be further classified into discrete and continuous data.

- ❖ **Discrete data:** This type of data is countable and can only take on specific values. Examples include the number of children in a family or the number of cars in a parking lot.
- ❖ **Continuous data:** This type of data is measurable and can take on any value within a range. Examples include height, weight, or temperature.

DATA COLLECTION METHODS

What are Data Collection Methods?

Data collection methods are techniques and procedures used to gather information for research purposes. These methods can range from simple self-reported surveys to more complex experiments and can involve either quantitative or qualitative approaches to data gathering. Some common data collection methods include surveys, interviews, observations, focus groups, experiments, and secondary data analysis. The data collected through these methods can then be analyzed and used to support or refute research hypotheses and draw conclusions about the study's subject matter.

Importance of Data Collection Methods

Data collection methods play a crucial role in the research process as they determine the quality and accuracy of the data collected. Here is some major importance of data collection methods.

- i. Determines the quality and accuracy of collected data.
- ii. Ensures that the data is relevant, valid, and reliable.
- iii. Helps reduce bias and increase the representativeness of the sample.
- iv. Essential for making informed decisions and accurate conclusions.
- v. Facilitates achievement of research objectives by providing accurate data.
- vi. Supports the validity and reliability of research findings.

TYPES OF DATA COLLECTION METHODS

The choice of data collection method depends on the research question being addressed, the type of data needed, and the resources and time available. You can categorize data collection methods into primary methods of data collection and secondary methods of data collection.

A) Primary Data Collection Methods

Primary data is collected from first-hand experience and is not used in the past. The data gathered by primary data collection methods are specific to the research's motive and highly accurate. Primary data collection methods can be divided into two categories: **quantitative methods** and **qualitative methods**.

i) Quantitative Methods

Quantitative data collection is the act of collecting numerical consumer insight data that translates easily into statistical or quantifiable information. Quantitative data collection methods include surveys, questionnaires, observations, and more. Further, this type of data collection typically involves a substantial number of respondents. This larger pool of people provides a large enough sample of extractable data to be statistically relevant for a more sizable population. While quantitative data is focused on extrapolating numerical data, the purpose of qualitative data is more about collecting contextual or descriptive information from a pool of people. In simpler terms, quantitative data is usually objective and answers questions about who, what, when, and where by looking at measurable numbers. However, qualitative data extracts more subjective information, such as why.

ii) Qualitative Methods

Qualitative data collection methods are especially useful in situations when historical data is not available. Or there is no need of numbers or mathematical calculations.

Qualitative research is closely associated with words, sounds, feeling, emotions, colors, and other elements that are non-quantifiable. These techniques are based on experience, judgment, intuition, conjecture, emotion, etc. Quantitative methods do not provide the motive behind participants' responses, often don't reach underrepresented populations, and span long periods to collect the data. Hence, it is best to combine quantitative methods with qualitative methods.

B) Secondary Data Collection Methods

Secondary data is the data that has been used in the past. The researcher can obtain data from the data sources, both internal and external, to the organizational data.

Internal sources of secondary data:

- Organization's health and safety records
- Mission and vision statements
- Financial Statements
- Magazines
- Sales Report
- CRM Software
- Executive summaries

External sources of secondary data:

- Government reports
- Press releases
- Business journals
- Libraries
- Internet

The secondary data collection methods can also involve quantitative and qualitative techniques. Secondary data is easily available and hence, less time-consuming and expensive than primary data. However, with the secondary data collection methods, the authenticity of the data gathered cannot be verified. The secondary data collection methods can also involve quantitative and qualitative observation techniques. Secondary data is easily available, less time-consuming, and more expensive than primary data. However, with the secondary data collection methods, the authenticity of the data gathered cannot be verified.

Regardless of the data collection method of your choice, there must be direct communication with decision-makers so that they understand and commit to acting according to the results.

WHAT ARE DATA COLLECTION TOOLS?

Data collection tools are essential instruments used to gather data in research that enable you to collect and store data from a variety of sources. They come in various forms, ranging from simple paper questionnaires to advanced software applications. Choosing the right data collection tool is crucial for obtaining accurate and reliable data. Researchers must consider the research question, the population being studied, and the desired level of detail when selecting the appropriate data collection tool. Each tool has its strengths and weaknesses, so the one that best suits their research objectives should be chosen.

There are a lot of different kinds of tools for collecting data, and the best one for a given situation depends on the user's specific data needs and goals.

Types of data collection tools

The success of market research depends on the accuracy of the data collected. However, it doesn't end here. It is important to analyze this data such that the management executives can make sense out of it.

There are multiple data collection and market research tools available in the market. Choosing the right one depends on your requirements. Some of the most commonly used data collection methods are:

a) Interviews

Interviews are a data collection method by which one person asks a series of questions to another who is part of the target audience. The one-to-one interview can be conducted in person or over the phone call. The answers are recorded and analyzed to understand their behavior, choices, and experiences. An interview gives high-quality of data as the respondent is less likely to give false responses when facing someone directly. The limitation of this mode of collecting data is that it is not feasible for large-scale data collection. Also, it is time-consuming to interview many people on a one-to-one basis. Interviews are most appropriate when the subject is complex or sensitive.

b) Observation

Observation is a data collection technique in which the target audience is given the problem statement, and their responses are noted with minimum interference from any moderator. While the respondent is busy thinking about answering the questions, their body language, mannerisms, facial expressions, voice tone, etc are noted. These non-verbal signs are an important input for making decisions. The best thing about this method is that it requires less preparation on the researcher's part. The downside is that the respondents might get conscious of themselves if somebody is constantly observing them. It might even make them feel uncomfortable. Observation is a data collection technique in which the target audience is given the problem statement, and their responses are noted with minimum interference from any moderator. While the respondent is busy thinking about answering the questions, their body language, mannerisms, facial expressions, voice tone, etc. are noted.

These non-verbal signs are an important input for making decisions. The best thing about this method is that it requires less preparation on the researcher's part.

The downside is that the respondents might get conscious of themselves if somebody is constantly observing them. It might even make them feel uncomfortable.

c) Surveys

Surveys are a list of questions designed in such a way that they touch upon various features of products and services. Based on the data collected, various analytical algorithms are run, and reports are generated. Surveys are conducted through online data collection tools by which the survey questions can be distributed through email, website, mobile app, QR code, or social media platforms. These data-gathering tools have the ability to generate various reports such as Trend Analysis, Data Segmentation, Correlation Analysis, Conjoint Analysis, etc. Often in large enterprises, multiple teams use the reports and so need to be shared with many people. This creates a need to export the reports in commonly used formats such as .xls, .doc, or .pdf. Moreover, the survey data can be used further by integrating third-party apps with the survey builder tool. This multiplies the usability of the data collected and improves ROI. Survey builder software comes with a lot of flexibility as the user can select from a variety of question types and add logic to customize the flow of survey based on the answers given by respondents. The respondents can answer based on their convenience. In case a respondent has started the survey but hasn't finished, the survey administrator can send scheduled reminders and increase the response rate. Surveys can reach a vast audience simultaneously and collect a lot of data in lesser time as compared to other methods.

d) Usage Data

With the growing use of technology, data is collected at various points from manufacture to delivery. This data can be used to improve business outcomes. Historical records, application usage data, sales reports, GPS location coordinates, etc., are some of the examples from which useful data can be gathered. Such data collection methods can be used when you want to optimize existing systems instead of creating new ones.

e) Focus Groups

Focus groups are a group of people who come from various fields of life. They are subjected to a group discussion in the presence of a moderator. As people come from different backgrounds, it leads to exploring various angles of thought. It is believed that diversity in thoughts leads to innovation. Focus groups help stakeholders to validate the facts on the spot and collect a diverse set of data. This method of data collection suffers from the risk of the presence of an over-influential person in the group. The moderator must be capable of managing the discussion and balancing the dynamics of the group. Focus groups are apt for beta testing of newly launched products. Data collection tools must be selected based on low barriers to entry, ability to scale and advanced features beyond basic data collection.

MEANING OF SAMPLE

A sample refers to a smaller, manageable version of a larger group. It is a subset containing the characteristics of a larger population. Samples are used in statistical testing when population sizes are too large for the test to include all possible members or observations. A sample should represent the population as a whole and not reflect any bias toward a specific attribute.

Understanding Samples

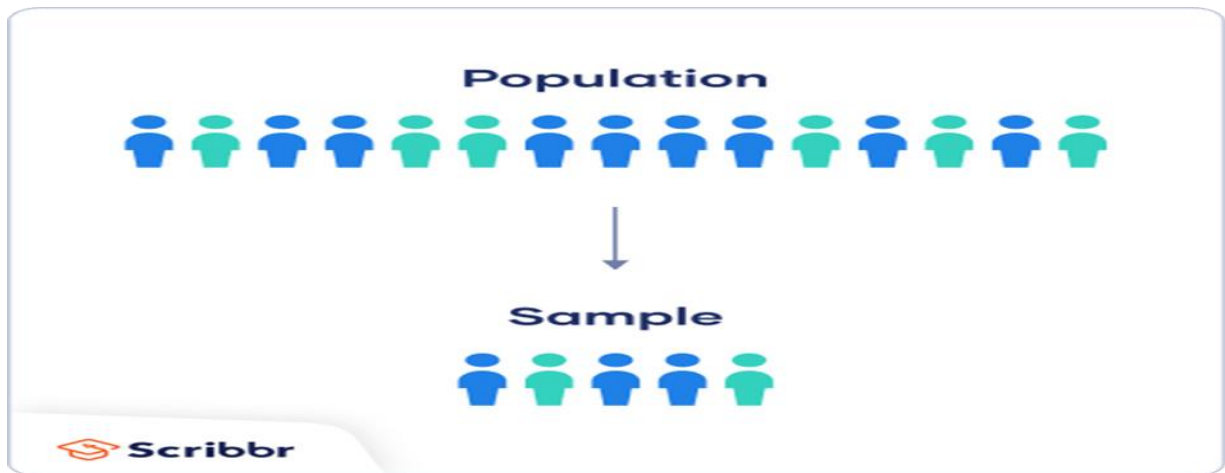
A sample is an unbiased number of observations taken from a population. In simple terms, a population is the total number of observations (i.e., individuals, animals, items, data, etc.) contained in a given group or context. A sample, in other words, is a portion, part, or fraction of the whole group, and acts as a subset of the population.

Population vs. sample

First, you need to understand the difference between a population and a sample, and identify the target population of your research.

The **population** is the entire group that you want to draw conclusions about. The population can be defined in terms of geographical location, age, income, or many other characteristics.

The **sample** is the specific group of individuals that you will collect data from.



It can be very broad or quite narrow: maybe you want to make inferences about the whole adult population of your country; maybe your research focuses on customers of a certain company, patients with a specific health condition, or students in a single school. It is important to carefully define your **target population** according to the purpose and practicalities of your project.

If the population is very large, demographically mixed, and geographically dispersed, it might be difficult to gain access to a representative sample. A lack of a representative sample affects the validity of your results, and can lead to several research biases, particularly sampling bias.

Sampling frame

The sampling frame is the actual list of individuals that the sample will be drawn from. Ideally, it should include the entire target population (and nobody who is not part of that population).

Example: Sampling frame “You are doing research on working conditions at a social media marketing company. Your population is all 1000 employees of the company. Your sampling frame is the company’s HR database, which lists the names and contact details of every employee”.

Sample size

The number of individuals you should include in your sample depends on various factors, including the size and variability of the population and your research design. There are different sample size calculators and formulas depending on what you want to achieve with statistical analysis.

However, samples are used in a variety of settings where research is conducted. Scientists, marketers, government agencies, economists, and research groups are among those who use samples for their studies and measurements.

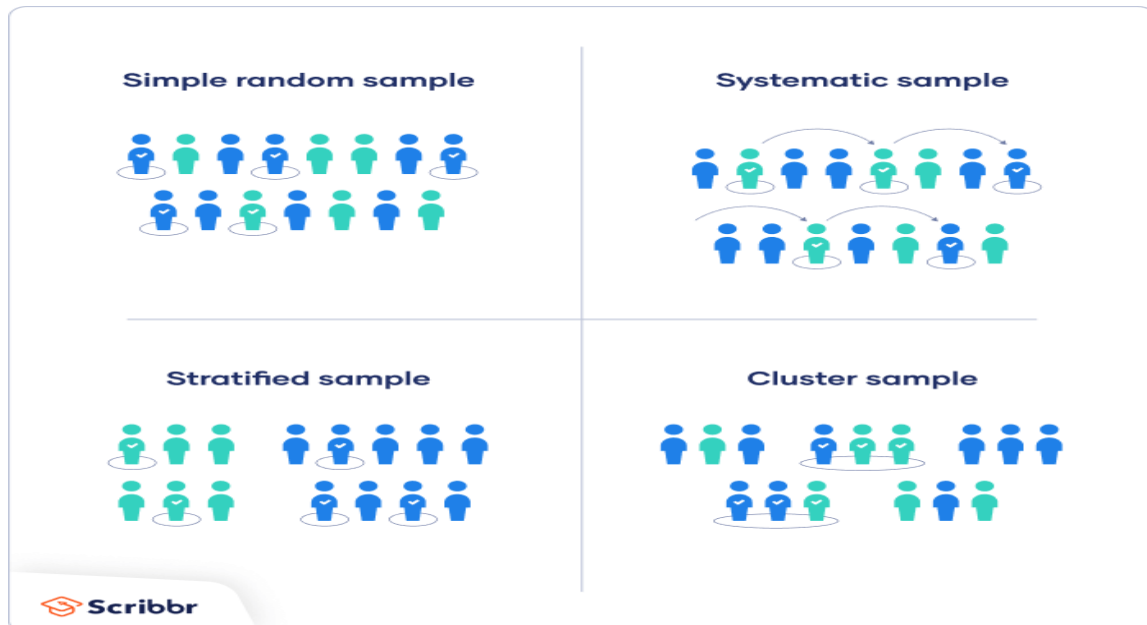
Using whole populations for research comes with challenges. Researchers may have problems gaining ready access to entire populations. And, because of the nature of some studies, researchers may have difficulties getting the results they need in a timely fashion. This is why people samples are used. Using a smaller number of people who represent the entire population can still produce valid results while reducing time and resources.

Samples used by researchers must resemble the broader population in order to make accurate inferences or predictions. All the participants in the sample should share the same characteristics and qualities.

So, if the study is about male college freshmen, the sample should be a small percentage of males that fit this description. Similarly, if a research group conducts a study on the sleep patterns of single women over 50, the sample should only include women within this demographic. There are two primary types of sampling methods that you can use in your research:

Probability sampling involves random selection, allowing you to make strong statistical inferences about the whole group. It means that every member of the population has a chance of being selected. It is mainly used in quantitative research. If you want to produce results that are

representative of the whole population, probability sampling techniques are the most valid choice. There are four main types of probability sample:



1. Simple random sampling

In a simple random sample, every member of the population has an equal chance of being selected. Your sampling frame should include the whole population. To conduct this type of sampling, you can use tools like random number generators or other techniques that are based entirely on chance.

Example: “You want to select a simple random sample of 1000 employees of a social media marketing company. You assign a number to every employee in the company database from 1 to 1000, and use a random number generator to select 100 numbers”.

2. Systematic sampling

Systematic sampling is similar to simple random sampling, but it is usually slightly easier to conduct. Every member of the population is listed with a number, but instead of randomly generating numbers, individuals are chosen at regular intervals.

Example: “All employees of the company are listed in alphabetical order. From the first 10 numbers, you randomly select a starting point: number 6. From number 6 onwards, every 10th person on the list is selected (6, 16, 26, 36, and so on), and you end up with a sample of 100 people”.

If you use this technique, it is important to make sure that there is no hidden pattern in the list that might skew the sample. For example, if the HR database groups employees by team, and team members are listed in order of seniority, there is a risk that your interval might skip over people in junior roles, resulting in a sample that is skewed towards senior employees.

3. Stratified sampling

Stratified sampling involves dividing the population into subpopulations that may differ in important ways. It allows you draw more precise conclusions by ensuring that every subgroup is properly represented in the sample.

To use this sampling method, you divide the population into subgroups (called strata) based on the relevant characteristic (e.g., gender identity, age range, income bracket, job role).

Based on the overall proportions of the population, you calculate how many people should be sampled from each subgroup. Then you use random or systematic sampling to select a sample from each subgroup.

Example: “The company has 800 female employees and 200 male employees. You want to ensure that the sample reflects the gender balance of the company, so you sort the population into two strata based on gender. Then you use random

sampling on each group, selecting 80 women and 20 men, which gives you a representative sample of 100 people”.

4. Cluster sampling

Cluster sampling also involves dividing the population into subgroups, but each subgroup should have similar characteristics to the whole sample. Instead of sampling individuals from each subgroup, you randomly select entire subgroups. If it is practically possible, you might include every individual from each sampled cluster. If the clusters themselves are large, you can also sample individuals from within each cluster using one of the techniques above. This is called multistage sampling.

This method is good for dealing with large and dispersed populations, but there is more risk of error in the sample, as there could be substantial differences between clusters. It's difficult to guarantee that the sampled clusters are really representative of the whole population.

Example: “The company has offices in 10 cities across the country (all with roughly the same number of employees in similar roles). You don't have the capacity to travel to every office to collect your data, so you use random sampling to select 3 offices – these are your clusters”.

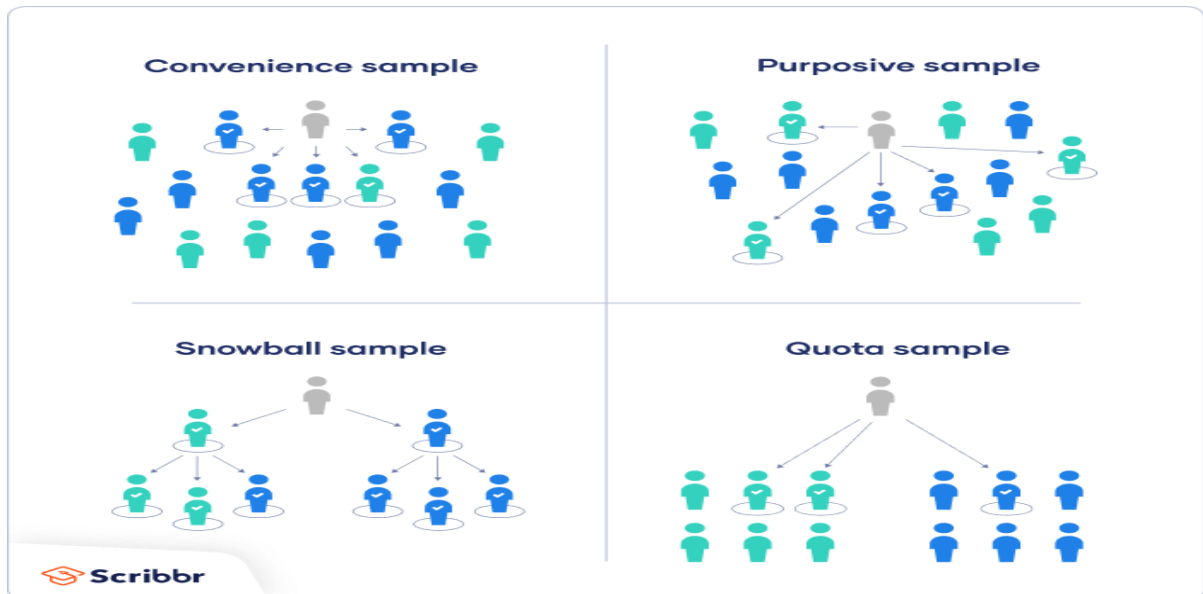
❖ **Non-probability sampling** involves non-random selection based on convenience or other criteria, allowing you to easily collect data.

In a non-probability sample, individuals are selected based on non-random criteria, and not every individual has a chance of being included.

This type of sample is easier and cheaper to access, but it has a higher risk of sampling bias. That means the inferences you can make about the population are weaker than with probability samples, and your conclusions

may be more limited. If you use a non-probability sample, you should still aim to make it as representative of the population as possible.

Non-probability sampling techniques are often used in exploratory and qualitative research. In these types of research, the aim is not to test a hypothesis about a broad population, but to develop an initial understanding of a small or under-researched population.



1. Convenience sampling

A convenience sample simply includes the individuals who happen to be most accessible to the researcher.

This is an easy and inexpensive way to gather initial data, but there is no way to tell if the sample is representative of the population, so it can't produce generalizable results. Convenience samples are at risk for both sampling bias and selection bias.

Example: “You are researching opinions about student support services in your university, so after each of your classes, you ask your fellow students to complete a survey on the topic. This is a convenient way to gather data, but as you only surveyed students taking the same classes as you at the same level, the sample is not representative of all the students at your university”.

2. Voluntary response sampling

Similar to a convenience sample, a voluntary response sample is mainly based on ease of access. Instead of the researcher choosing participants and directly contacting them, people volunteer themselves (e.g., by responding to a public online survey).

Voluntary response samples are always at least somewhat biased, as some people will inherently be more likely to volunteer than others, leading to self-selection bias.

Example: “You send out the survey to all students at your university and a lot of students decide to complete it. This can certainly give you some insight into the topic, but the people who responded are more likely to be those who have strong opinions about the student support services, so you can’t be sure that their opinions are representative of all students”.

3. Purposive sampling

This type of sampling, also known as judgement sampling, involves the researcher using their expertise to select a sample that is most useful to the purposes of the research.

It is often used in qualitative research, where the researcher wants to gain detailed knowledge about a specific phenomenon rather than make statistical inferences, or where the population is very small and specific. An effective purposive sample must have clear criteria and rationale for inclusion. Always make sure to describe your inclusion and exclusion criteria and beware of observer bias affecting your arguments.

Example: “You want to know more about the opinions and experiences of disabled students at your university, so you purposefully select a number of students with different support needs in order to gather a varied range of data on their experiences with student services”.

4. Snowball sampling

If the population is hard to access, snowball sampling can be used to recruit participants via other participants. The number of people you have access to “snowballs” as you get in contact with more people. The downside here is also representativeness, as you have no way of knowing how representative your sample is due to the reliance on participants recruiting others. This can lead to sampling bias.

Example: “You are researching experiences of homelessness in your city. Since there is no list of all homeless people in the city, probability sampling isn’t possible. You meet one person who agrees to participate in the research, and she puts you in contact with other homeless people that she knows in the area”.

5. Quota sampling

Quota sampling relies on the non-random selection of a predetermined number or proportion of units. This is called a quota.

You first divide the population into mutually exclusive subgroups (called strata) and then recruit sample units until you reach your quota. These units share specific characteristics, determined by you prior to forming your strata. The aim of quota sampling is to control what or who makes up your sample.

Example: “You want to gauge consumer interest in a new produce delivery service in Boston, focused on dietary preferences. You divide the population into meat eaters, vegetarians, and vegans, drawing a sample of 1000 people. Since the company wants to cater to all consumers, you set a quota of 200 people for each dietary group. In this way, all dietary preferences are equally represented in your research, and you can easily compare these groups. You continue recruiting until you reach the quota of 200 participants for each subgroup.

SAMPLING PROCESS

The sampling process involves five stages, which include:

1. **Defining the target population:** This stage involves defining the population that the researcher wants to study.
2. **Choosing the sampling frame:** This stage involves identifying the list of individuals or elements that make up the population.
3. **Selecting the sampling technique:** This stage involves choosing the appropriate sampling technique based on the research question, the type of data needed, and the resources and time available.
4. **Determining the sample size:** This stage involves deciding on the number of individuals or elements that will be included in the sample.
5. **Selecting the sample:** This stage involves selecting the individuals or elements that will be included in the sample, using the chosen sampling technique.

MODULE THREE

SOURCES OF RESEARCH IDEAS

1. ***Replicating, clarifying, or challenging previous research:*** Researchers often find inspiration in the work of others, such as resolving conflicting results or building upon existing knowledge.
2. ***New technology:*** The impact of new technologies, such as social media platforms, can also spark research ideas’
3. ***Serendipity:*** Surprise findings that researchers want to explore further can lead to new research ideas.
4. ***Anomalies:*** Unexpected situations that should not technically exist can also be a source of research inspiration.
5. ***Common sense research:*** Sometimes, research ideas come from challenging long-held beliefs or exploring further something that is commonly accepted.
6. ***Problems or concerns in a specific field:*** Research in applied fields often stems from a problem or concern that needs to be addressed, such as a goal that an agency is trying to achieve or a policy change.
7. ***Informal observations:*** Everyday experiences and encounters can provide inspiration for research ideas.
8. ***Consulting with experienced researchers:*** Novice researchers can find inspiration by consulting with more experienced researchers or by reading professional journals and abstracts.
9. ***Systematic sources:*** Study and knowledge of a specific topic, including the results of past research, theories, and classroom lectures, can form the foundation for systematic sources of research ideas.

STATEMENT OF THE PROBLEM

The basics of writing a statement of the problem for your research proposal

A *statement of the problem* is used in research work as a claim that outlines the problem addressed by a study. The statement of the problem briefly addresses the question: What is the problem that the research will address?

What are the goals of a statement of the problem?

The ultimate goal of a statement of the problem is to transform a generalized problem (something that bothers you; a perceived lack) into a targeted, well-defined problem; one that can be resolved through focused research and careful decision-making.

Writing a statement of the problem should help you clearly identify the purpose of the research project you will propose. Often, the statement of the problem will also serve as the basis for the introductory section of your final proposal, directing your reader's attention quickly to the issues that your proposed project will address and providing the reader with a concise statement of the proposed project itself.

A statement of problem need not be long and elaborate: one page is more than enough for a good statement of problem.

What are the key characteristics of a statement of the problem?

A good research problem should have the following characteristics:

- i. It should address a gap in knowledge.
- ii. It should be significant enough to contribute to the existing body of research
- iii. It should lead to further research
- iv. The problem should render itself to investigation through collection of data

- v. It should be of interest to the researcher and suit his/her skills, time, and resources
- vi. The approach towards solving the problem should be ethical
- vii. What is the format for writing a statement of the problem?

A persuasive statement of problem is usually written in three parts:

Part A (The ideal): Describes a desired goal or ideal situation;

Explains how things should be.

Part B (The reality): Describes a condition that prevents the goal, state, or value in Part A from being achieved or realized at this time; explains how the current situation falls short of the goal or ideal.

Part C (The consequences): Identifies the way you propose to improve the current situation and move it closer to the goal or ideal.

Here is an example:

Statement 1

The Ministry of Youth is dedicated to allocating enterprise development funds to both the youth and women. These funds are made available in order to start entrepreneurial ventures that create and expand employment. (Provide relevant statistics and quote)

Statement 2

One of the main focuses of the ministry is consistency. Unfortunately, consistency in allocating funds to the next generation of recipients requires prior knowledge of previous allocations and established practices. The current continuous disbursement method does not allow for adequate analysis of previous disbursements before a current disbursement is done.

Statement 3

Continuing with this current disbursement method prevents consistency and causes decisions to become grossly political, which in turn inhibits the achievement of the goals of the funds. Developing a more informed disbursement system could help better implement the consistency focus of the ministry and at the same time help the ministry better monitor and evaluate its funds.

LITERATURE REVIEW

A literature review surveys prior research published in books, scholarly articles, and any other sources relevant to a particular issue, area of research, or theory, and by so doing, provides a description, summary, and critical evaluation of these works in relation to the research problem being investigated. Literature reviews are designed to provide an overview of sources you have used in researching a particular topic and to demonstrate to your readers how your research fits within existing scholarship about the topic.

It is important to think of knowledge in a given field as consisting of three layers. First, there are the **primary studies** that researchers conduct and publish. Second are the **reviews of those studies** that summarize and offer new interpretations built from and often extending beyond the primary studies. Third, there are the **perceptions, conclusions, opinion, and interpretations** that are shared informally among scholars that become part of the body of epistemological traditions within the field.

In composing a literature review, it is important to note that it is often this third layer of knowledge that is cited as "true" even though it often has only a loose relationship to the primary studies and secondary literature reviews. Given this, while literature reviews are designed to provide an overview and synthesis of pertinent sources you have explored, there are a number of

approaches you could adopt depending upon the type of analysis underpinning your study.

The analytical features of a literature review

- Give a new interpretation of old material or combine new with old interpretations,
- Trace the intellectual progression of the field, including major debates,
- Depending on the situation, evaluate the sources and advise the reader on the most pertinent or relevant research, or
- Usually in the conclusion of a literature review, identify where gaps exist in how a problem has been researched to date.

The purpose of a literature review

- Place each work in the context of its contribution to understanding the research problem being studied.
- Describe the relationship of each work to the others under consideration.
- Identify new ways to interpret prior research.
- Reveal any gaps that exist in the literature.
- Resolve conflicts amongst seemingly contradictory previous studies.
- Identify areas of prior scholarship to prevent duplication of effort.
- Point the way in fulfilling a need for additional research.
- Locate your own research within the context of existing literature [very important].

Types of Literature Reviews

1. Argumentative Review

This form examines literature selectively in order to support or refute an argument, deeply embedded assumption, or philosophical problem already established in the literature. The purpose is to develop a body of literature that establishes a contrarian viewpoint. However, note that they can also introduce problems of bias when they are used to make summary claims of the sort found in systematic reviews.

2. Integrative Review

Considered a form of research that reviews, critiques, and synthesizes representative literature on a topic in an integrated way such that new frameworks and perspectives on the topic are generated. The body of literature includes all studies that address related or identical hypotheses or research problems. A well-done integrative review meets the same standards as primary research in regard to clarity, rigor, and replication. This is the most common form of review in the social sciences.

3. Historical Review

Historical literature reviews focus on examining research throughout a period of time, often starting with the first time an issue, concept, theory, phenomena emerged in the literature, then tracing its evolution within the scholarship of a discipline. The purpose is to place research in a historical context to show familiarity with state-of-the-art developments and to identify the likely directions for future research.

4. Methodological Review

A review does not always focus on **what** someone said [findings], but **how** they came about saying what they say [method of analysis]. Reviewing methods of analysis provides a framework of understanding at different levels [i.e., those of theory, substantive fields, research approaches, and data collection and analysis techniques], how researchers draw upon a wide variety of knowledge ranging from the conceptual level to practical documents for use in fieldwork in the areas of ontological and epistemological consideration, quantitative and qualitative integration, sampling, interviewing, data collection, and data analysis. This approach helps highlight ethical issues which you should be aware of and consider as you go through your own study.

5. Systematic Review

This form consists of an overview of existing evidence pertinent to a clearly formulated research question, which uses pre-specified and standardized methods to identify and critically appraise relevant research, and to collect, report, and analyze data from the studies that are included in the review. The goal is to deliberately document, critically evaluate, and summarize scientifically all of the research about a clearly defined research problem. Typically, it focuses on a very specific empirical question, often posed in a cause-and-effect form, such as "To what extent does A contribute to B?" This type of literature review is primarily applied to examining prior research studies in clinical medicine and allied health fields, but it is increasingly being used in the social sciences.

6. Theoretical Review

The purpose of this form is to examine the corpus of theory that has accumulated in regard to an issue, concept, theory, phenomena. The theoretical literature review helps to establish what theories already exist, the relationships between them, to what degree the existing theories have been investigated, and to develop new hypotheses to be tested. Often this form is used to help establish a lack of appropriate theories or reveal that current theories are inadequate for explaining new or emerging research problems. The unit of analysis can focus on a theoretical concept or a whole theory or framework.

NOTE: Most often the literature review will incorporate some combination of types. [For example, a review that examines literature supporting or refuting an argument, assumption, or philosophical problem related to the research problem will also need to include writing supported by sources that establish the history of these arguments in the literature.](#)

Development of the Literature Review

Factors to be considered before writing the literature review:

- Clarify
- Find Models
- Narrow the Topic
- Consider Whether Your Sources are Current

Ways to Organize Your Literature Review

1. Chronology of Events

If your review follows the chronological method, you could write about the materials according to when they were published. This approach should only be followed if a clear path of research building on previous research can be identified and that these trends follow a clear chronological order of development. For example, a literature review that focuses on continuing research about the emergence of German economic power after the fall of the Soviet Union.

2. By Publication

Order your sources by publication chronology, then, only if the order demonstrates a more important trend. For instance, you could order a review of literature on environmental studies of brown fields if the progression revealed, for example, a change in the soil collection practices of the researchers who wrote and/or conducted the studies.

3. Thematic [“conceptual categories”]

A thematic literature review is the most common approach to summarizing prior research in the social and behavioural sciences. Thematic reviews are organized around a topic or issue, rather than the progression of time, although the progression of time may still be incorporated into a thematic review. For example, a review of the Internet’s impact on American presidential politics could focus on the development of online political satire.

Writing Your Literature Review

Once you've settled on how to organize your literature review, you're ready to write each section. When writing your review, keep in mind these issues.

- **Use Evidence**

A literature review section is, in this sense, just like any other academic research paper. Your interpretation of the available sources must be backed up with evidence [citations] that demonstrates that what you are saying is valid.

- **Be Selective**

Select only the most important points in each source to highlight in the review. The type of information you choose to mention should relate directly to the research problem, whether it is thematic, methodological, or chronological. Related items that provide additional information, but that are not key to understanding the research problem, can be included in a list of further readings.

- **Use Quotes Sparingly**

Some short quotes are appropriate if you want to emphasize a point, or if what an author stated cannot be easily paraphrased. Sometimes you may need to quote certain terminology that was coined by the author, is not common knowledge, or taken directly from the study. Do not use extensive quotes as a substitute for using your own words in reviewing the literature.

- **Summarize and Synthesize**

Remember to summarize and synthesize your sources within each thematic paragraph as well as throughout the review. Recapitulate important features of a research study, but then synthesize it by rephrasing the study's significance and relating it to your own work and the work of others.

- **Keep Your Own Voice**

While the literature review presents others' ideas, your voice [the writer's] should remain front and center. For example, weave references to other sources into what you are writing but maintain your own voice by starting and ending the paragraph with your own ideas and wording.

- **Use Caution When Paraphrasing**

When paraphrasing a source that is not your own, be sure to represent the author's information or opinions accurately and in your own words. Even when paraphrasing an author's work, you still must provide a citation to that work.

References

Fink, Arlene. *Conducting Research Literature Reviews: From the Internet to Paper*. Fourth edition. Thousand Oaks, CA: SAGE, 2014.

Sutton, Anthea. *Systematic Approaches to a Successful Literature Review*. Los Angeles, CA: Sage Publications, 2016.

Cook, Kathleen E. and Elise Murowchick. "Do Literature Review Skills Transfer from One Course to Another?" *Psychology Learning and Teaching* 13 (March 2014): 3-11

RESEARCH QUESTION

A research question is an inquiry that the research attempts to answer. It is the heart of the systematic investigation. Research questions are the most important step in any research project. In essence, it initiates the research project and establishes the pace for the specific research

A research question must be:

- **Clear:** It provides enough detail that the audience understands its purpose without any additional explanation.
- **Focused:** It is so specific that it can be addressed within the time constraints of the writing task.
- **Succinct:** It is written in the shortest possible words.
- **Complex:** It is not possible to answer it with a "yes" or "no", but requires analysis and synthesis of ideas before somebody can create a solution.
- **Argumental:** Its potential answers are open for debate rather than accepted facts.

A good research question usually focuses on the research and determines the research design, methodology, and hypothesis. It guides all phases of inquiry, data collection, analysis, and reporting. You should gather valuable information by asking the right questions.

TYPES OF RESEARCH QUESTIONS

There must be research questions for every type of research. Your research question will be based on the type of research you want to conduct and the type of data collection.

The first step in designing research involves identifying a gap and creating a focused research question.



Steps to Write Research Questions



The following are the steps to follow in formulating research questions:

a) Select an interesting topic:

Always choose a topic that interests you. Because if your curiosity isn't aroused by a subject, you'll have a hard time conducting research around it. Also, it's better that you pick something that's neither too narrow or too broad.

b) Do preliminary research on the topic:

Search for relevant literature to gauge what problems have already been tackled by scholars. You can do that conveniently through repositories like Scispace, where you'll find millions of papers in one place. Once you do find the papers you're looking for, try our reading assistant, SciSpace Copilot to get simple explanations for the paper. You'll be able to quickly understand the abstract, find the key takeaways, and the main arguments presented in the paper. This will give you a more contextual understanding of your subject and you'll have an easier time identifying knowledge gaps in your discipline.

c) Consider your audience

It is essential to understand your audience to develop focused research questions for essays or dissertations. When narrowing down your topic, you can identify aspects that might interest your audience.

d) Ask questions

Asking questions will give you a deeper understanding of the topic. Evaluate your question through the What, Why, When, How, and other open-ended questions assessment.

e) Assess your question

Once you have created a research question, assess its effectiveness to determine if it is useful for the purpose. Refine and revise the dissertation research question multiple times.

The following are characteristics of good research question:

Researchable

- It should only cover one issue.
- A subjective judgment should not be included in the question.
- It can be answered with data analysis and research.

Specific and Practical

- It should not contain a plan of action, policy, or solution.
- It should be clearly defined
- Within research limits

Complex and Arguable

- It shouldn't be difficult to answer.
- To find the truth, you need in-depth knowledge
- Allows for discussion and deliberation

Original and Relevant

- It should be in your area of study
- Its results should be measurable
- It should be original

RESEARCH OBJECTIVES

Research objectives describe what your research is trying to achieve and explain why you are pursuing it. They summarize the approach and purpose of your project and help to focus your research. They should guide every step of the research process, including how you collect data, build your argument, and develop your conclusions.

Your objectives should appear in the introduction of your **research paper**, and at the end of your **problem statement**. They should:

- Establish the **scope** and depth of your project
- Contribute to your **research design**
- Indicate how your project will contribute to existing knowledge

RESEARCH AIMS/GENERAL OBJECTIVE

A distinction is often made between research objectives and research aims. A research aim typically refers to **a broad statement indicating the general purpose of your research project**. It should appear at the end of your problem statement, before your research objectives.

IMPORTANT OF RESEARCH OBJECTIVES

Research objectives are important because they:

- **Establish the scope and depth of your project:** This helps you avoid unnecessary research. It also means that your research methods and conclusions can easily be evaluated.
- **Contribute to your research design:** When you know what your objectives are, you have a clearer idea of what methods are most appropriate for your research.

- **Indicate how your project will contribute to extant research:** They allow you to display your knowledge of up-to-date research, employ or build on current research methods, and attempt to contribute to recent debates.

HOW TO WRITE RESEARCH AIMS AND OBJECTIVES

Once you've established a research problem you want to address, you need to decide **how** you will address it. This is where your research aims and objectives come in.

Step 1: Decide on a general aim

Your research aim should reflect your research problem and should be relatively broad.

Example: Research aim “To assess the safety features and response times of self-driving cars”.

Step 2: Decide on specific objectives

Break down your aim into a limited number of steps that will help you resolve your research problem. What specific aspects of the problem do you want to examine or understand?

Example: Research objective “To measure the response time of self-driving cars to pedestrians”.

Step 3: Formulate your aims and objectives

Once you've established your research aim and objectives, you need to explain them clearly and concisely to the reader. You'll lay out your aims and objectives at the end of your problem statement, which appears in your introduction. Frame

them as clear declarative statements, and use appropriate verbs to accurately characterize the work that you will carry out.

Example: Explaining your research aim and objectives “The aim of this research is to **assess** the safety features and response times of self-driving cars. Using quantitative methods, I will **measure** the response time of self-driving cars, first to other moving vehicles and then to pedestrians. I will also **test** their awareness of nearby inanimate obstacles (e.g., sidewalks)”.

CHARACTERISTICS OF RESEARCH OBJECTIVES

The acronym “SMART” is commonly used in relation to research objectives. It states that your objectives should be:

- **Specific:** Make sure your objectives aren’t overly vague. Your research needs to be clearly defined in order to get useful results.
- **Measurable:** Know how you’ll measure whether your objectives have been achieved.
- **Achievable:** Your objectives may be challenging, but they should be feasible. Make sure that relevant groundwork has been done on your topic or that relevant primary or secondary sources exist. Also ensure that you have access to relevant research facilities (labs, library resources, research databases, etc.).
- **Relevant:** Make sure that they directly address the research problem you want to work on and that they contribute to the current state of research in your field.
- **Time-based:** Set clear deadlines for objectives to ensure that the project stays on track.

Ryan, E. (2023, May 31). *Research Objectives | Definition & Examples*. Scribbr. Retrieved October 25, 2023, from <https://www.scribbr.com/research-process/research-objectives/>

SCOPE OF THE RESEARCH

The scope of the study explains the extent to which the research area will be explored in the work and specifies the parameters within the study will be operating. Basically, this means that you will have to define what the study is going to cover and what it is focusing on. Similarly, you also have to define what the study is not going to cover. This will come under the limitations.

Typically, the information you need to include in the scope of the study would cover the following:

- General purpose of the study
- The population or sample that you are studying
- The duration of the study
- The topics or theories that you will discuss
- The geographical location covered in the study

SIGNIFICANCE OF THE RESEARCH

The significance of the study is a written statement that explains why your research is needed. It is a justification of the importance of your work and impact it has on your research field, its contribution to new knowledge and how others will benefit from it.

How do you write the significance of the study?

Significance of the research and thesis study is written from two perspectives:

1. What is the general contribution of your research on your field?
2. What is specific contribution have the researcher made to the knowledge and who does this benefit the most.

RESEARCH VARIABLES

Shukla (2018) says that variable is concerned with variation in presence of something in person, object, animal, and situation or in any natural phenomena. It can be defined as a characteristic under study of which an identity or value changes or is possible to change. Similarly, variable is concerned with variation in presence of something in person, object, animal, and situation or in any natural phenomena.

TYPES OF VARIABLES

Independent variables

An independent variable is a singular characteristic that the other variables in your experiment cannot change. [Age](#) is an example of an independent variable. Where someone lives, what they eat or how much they exercise are not going to change their age. Independent variables can, however, change other variables. In studies, researchers often try to find out whether an independent variable causes other variables to change and in what way.

[Example: In a study investigating the impact of different teaching methods on student performance, the independent variable might be the teaching method \(e.g., traditional lecture, active learning, online instruction\).](#)

Dependent variables

A dependent variable relies on and can be changed by other components. A [grade](#) on an exam is an example of a dependent variable because it depends on factors such as how much sleep you got and how long you studied. Independent variables can influence dependent variables, but dependent variables cannot influence independent variables.

[Example: In the same teaching method study, the dependent variable might be student performance, such as test scores or grades.](#)

Extraneous variable

Extraneous variables are factors that affect the dependent variable but that the researcher did not originally consider when designing the experiment. These unwanted variables can unintentionally change a study's results or how a researcher interprets those results. Take, for example, a study assessing whether private tutoring or online courses are more effective at improving students' Spanish test scores.

Extraneous variables that might unintentionally influence the outcome include parental support, prior knowledge of a foreign language or socioeconomic status.

Criteria in variable selection

- Being consistent with the goal
- Being measurable
- Being affordable and can be fitted with the study design
- Being prevalent and common in the community
- Being reliable (i.e., produces stable and consistent results over a period of time)
- Being valid (e.g., kg scale is not a suitable scale to measure the

- height of the people)
- Can be measured using available tools
- Not being time-consuming
- Not being out of research scope Importance of the variables
- They help to measure concepts in a study
- Allow researchers to draw conclusions about cause-and-effect
- Relationships.
- They are basic units of the information studied and interpreted in research studies – researchers carefully analyse and interpret the value(s) of each variable to make sense of how things relate to each other in descriptive study or what has happened in an experiment.
- They drive the research process

RESEARCH HYPOTHESIS

What is Hypothesis?

Hypothesis is an assumption or an idea proposed for the sake of argument so that it can be tested. It is a precise, testable statement of what the researchers predict will be outcome of the study. Hypothesis usually involves proposing a relationship between two variables: the independent variable (what the researchers change) and the dependent variable (what the research measures).

What is a Research Hypothesis?

Research hypothesis is a statement that introduces a research question and proposes an expected result. It is an integral part of the scientific method that forms the basis of scientific experiments. Therefore, you need to be careful and thorough when building your research hypothesis. A minor flaw in the construction of your hypothesis could have an adverse effect on your experiment. In research, there is a convention that the hypothesis is written in two forms, the null hypothesis, and the alternative hypothesis (called the experimental hypothesis when the method of investigation is an experiment).

Separating a Hypothesis from a Prediction

Outside of academia, hypothesis and prediction are often used interchangeably. In research writing, this is not only confusing but also incorrect. And although a hypothesis and prediction are guesses at their core, there are many differences between them.

- Hypothesis is an educated guess.
- Hypothesis is a testable prediction validated through research.
- It aims to analyze the gathered evidence and facts to define a relationship between variables and put forth a logical explanation behind the nature of events.

WHILE

- Prediction is assumption or expected outcome made without any backing evidence.
- It is more fictionally inclined regardless of where they originate from.

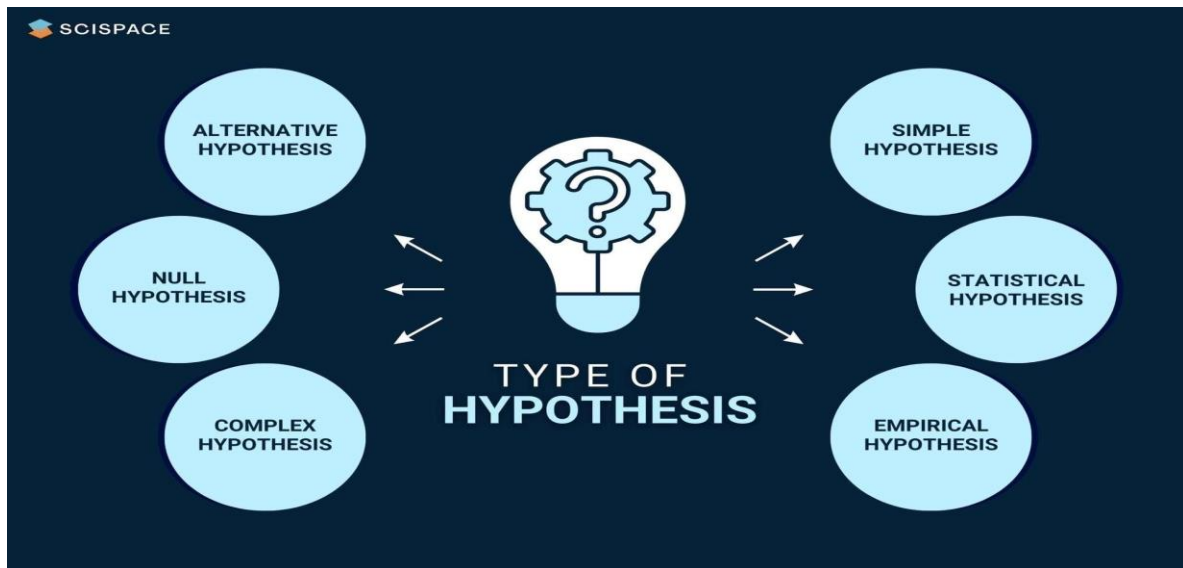
From the above reasons, hypothesis holds much more weight than a prediction. It sticks to the scientific method rather than pure guesswork. For example "Planets revolve around the Sun." is an example of a hypothesis as it is previous knowledge and observed trends. Additionally, we can test it through the scientific method.

Whereas "COVID-19 will be eradicated by 2030." is a prediction. Even though it results from past trends, we can't prove or disprove it. So, the only way this gets validated is to wait and watch if COVID-19 cases end by 2030.

TYPES OF HYPOTHESIS

Some would stand by the notion that there are only two types of hypotheses: a **Null hypothesis** and an **Alternative hypothesis**. While that may have some truth to it, it would be better to fully distinguish the most common forms as these terms come up so often, which might leave you out of context.

Apart from Null and Alternative, there are **Complex, Simple, Directional, Non-Directional, Statistical, and Associative and casual hypotheses**. They don't necessarily have to be exclusive, as one hypothesis can tick many boxes, but knowing the distinctions between them will make it easier for you to construct your own.



1. Null hypothesis

A null hypothesis proposes no relationship between two variables. Denoted by H_0 , it is a negative statement like “Attending physiotherapy sessions does not affect athletes' on-field performance.” Here, the author claims physiotherapy sessions have no effect on on-field performances. Even if there is, it's only a coincidence.

2. Alternative hypothesis

Considered to be the opposite of a null hypothesis, an alternative hypothesis is denoted as H_1 or H_a . It explicitly states that the dependent variable affects the independent variable. A good alternative hypothesis example is “Attending physiotherapy sessions improves athletes' on-field performance.” or “Water evaporates at 100°C ”.

The alternative hypothesis further branches into directional and non-directional.

- **Directional hypothesis:** A hypothesis that states the result would be either positive or negative is called directional hypothesis. It accompanies H1 with either the '<' or '>' sign.
- **Non-directional hypothesis:** A non-directional hypothesis only claims an effect on the dependent variable. It does not clarify whether the result would be positive or negative. The sign for a non-directional hypothesis is '≠.'

3. Simple hypothesis

A simple hypothesis is a statement made to reflect the relation between exactly two variables. One independent and one dependent. Consider the example, "Smoking is a prominent cause of lung cancer." The dependent variable, lung cancer, is dependent on the independent variable, smoking.

4. Complex hypothesis

In contrast to a simple hypothesis, a complex hypothesis implies the relationship between multiple independent and dependent variables. For instance, "Individuals who eat more fruits tend to have higher immunity, lesser cholesterol, and high metabolism." The independent variable is eating more fruits, while the dependent variables are higher immunity, lesser cholesterol, and high metabolism.

5. Associative and casual hypothesis

Associative and casual hypotheses don't exhibit how many variables there will be. They define the relationship between the variables. In an associative hypothesis, changing any one variable, dependent or independent, affects others. In a casual hypothesis, the independent variable directly affects the dependent.

6. Empirical hypothesis

Also referred to as the working hypothesis, an empirical hypothesis claims a theory's validation via experiments and observation. This way, the statement appears justifiable and different from a wild guess. Say, the hypothesis is “Women who take iron tablets face a lesser risk of anemia than those who take vitamin B12.” This is an example of an empirical hypothesis where the researcher the statement after assessing a group of women who take iron tablets and charting the findings.

7. Statistical hypothesis

The point of a statistical hypothesis is to test an already existing hypothesis by studying a population sample. Hypothesis like “44% of the Indian population belong in the age group of 22-27.” leverage evidence to prove or disprove a particular statement.

Characteristics of a Good Hypothesis

There are a number of considerations to keep in mind when constructing a hypothesis, as they are important for valid verification. The wording of a hypothesis therefore must have certain attributes that make it easier for you to ascertain its validity. These attributes are:

- Clearly formulated and free of language errors and ambiguity
- Concise and not unnecessarily verbose

- Logical and consistent with the current understanding of the subject
- Relevant to the research question and objectives
- Feasible to test within the scope of the study
- Reflects the population or sample being studied

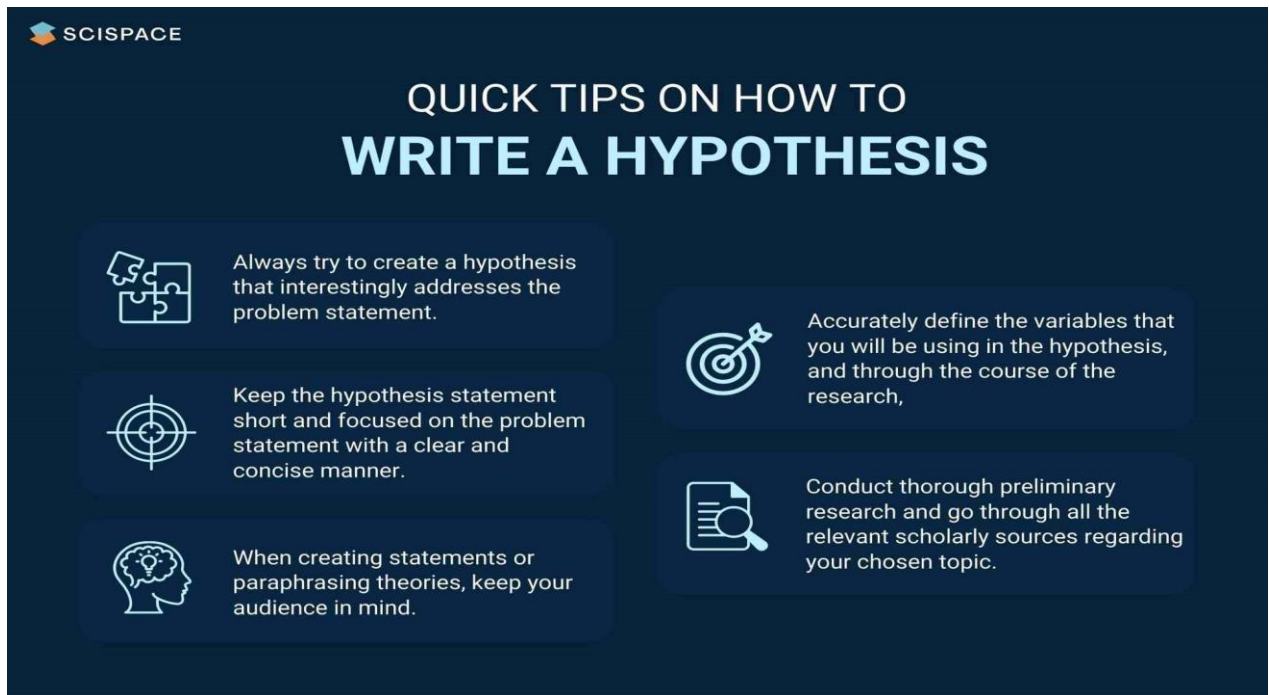
Significance of research hypotheses

- Provide a clear direction for the study
- Guide the selection of appropriate research methods and data analysis techniques
- Facilitate the interpretation of research findings
- Contribute to the development of scientific knowledge and understanding

References

<https://www.enago.com/academy/how-to-develop-a-good-research-hypothesis/>

How to Write a Hypothesis



SCISPACE

QUICK TIPS ON HOW TO WRITE A HYPOTHESIS

-  Always try to create a hypothesis that interestingly addresses the problem statement.
-  Keep the hypothesis statement short and focused on the problem statement with a clear and concise manner.
-  When creating statements or paraphrasing theories, keep your audience in mind.
-  Accurately define the variables that you will be using in the hypothesis, and through the course of the research,
-  Conduct thorough preliminary research and go through all the relevant scholarly sources regarding your chosen topic.

DATA ANALYSIS

This is the process of systematically applying statistical and/or logical techniques to describe and illustrate, condense and recap, and evaluate data. While data analysis in qualitative research can include statistical procedures, many times analysis becomes an ongoing iterative process where data is continuously collected and analyzed almost simultaneously. Indeed, researchers generally analyze for patterns in observations through the entire data collection phase. The form of the analysis is determined by the specific qualitative approach taken (field study, ethnography content analysis, oral history, biography, unobtrusive research) and the form of the data (field notes, documents, audiotape, videotape).

An essential component of ensuring data integrity is the accurate and appropriate analysis of research findings. Improper statistical analyses distort scientific findings, mislead casual readers, and may negatively influence the public

perception of research. Integrity issues are just as relevant to analysis of non-statistical data as well.

Issues to be considered in data analysis

There are a number of issues that researchers should be cognizant of with respect to data analysis. These include:

- Having the necessary skills to analyze
- Concurrently selecting data collection methods and appropriate analysis
- Drawing unbiased inference
- Inappropriate subgroup analysis
- Following acceptable norms for disciplines
- Determining statistical significance
- Lack of clearly defined and objective outcome measurements
- Providing honest and accurate analysis
- Manner of presenting data
- Environmental/contextual issues
- Data recording method
- Partitioning 'text' when analyzing qualitative data
- Training of staff conducting analyses
- Reliability and Validity
- Extent of analysis

TYPES OF DATA ANALYSIS METHODS

a) Descriptive analysis - What happened.

The descriptive analysis method is the starting point for any analytic reflection, and it aims to answer the question of what happened? It does this by ordering, manipulating, and interpreting raw data from various sources to turn it into valuable insights for your organization.

Performing descriptive analysis is essential, as it enables us to present our insights in a meaningful way. Although it is relevant to mention that this analysis on its own will not allow you to predict future outcomes or tell you the answer to questions like why something happened, it will leave your data organized and ready to conduct further investigations.

b) Exploratory analysis - How to explore data relationships.

As its name suggests, the main aim of the exploratory analysis is to explore. Prior to it, there is still no notion of the relationship between the data and the variables. Once the data is investigated, exploratory analysis helps you to find connections and generate hypotheses and solutions for specific problems. A typical area of application for it is data mining.

c) Diagnostic analysis - Why it happened.

Diagnostic data analytics empowers analysts and executives by helping them gain a firm contextual understanding of why something happened. If you know why something happened as well as how it happened, you will be able to pinpoint the exact ways of tackling the issue or challenge.

Designed to provide direct and actionable answers to specific questions, this is one of the world's most important methods in research, among its other key organizational functions such as retail analytics, e.g.

c) Predictive analysis - What will happen.

The predictive method allows you to look into the future to answer the question: what will happen? In order to do this, it uses the results of the previously mentioned descriptive, exploratory, and diagnostic analysis, in addition to machine learning (ML) and artificial intelligence (AI). Through this, you can uncover future trends, potential problems or inefficiencies, connections, and casualties in your data.

With predictive analysis, you can unfold and develop initiatives that will not only enhance your various operational processes but also help you gain an all-important edge over the competition. If you understand why a trend, pattern, or event happened through data, you will be able to develop an informed projection of how things may unfold in particular areas of the business.

e) Prescriptive analysis - How will it happen.

Another of the most effective types of analysis methods in research. Prescriptive data techniques cross over from predictive analysis in the way that it revolves around using patterns or trends to develop responsive, practical business strategies.

By drilling down into prescriptive analysis, you will play an active role in the data consumption process by taking well-arranged sets of visual data and using it as a powerful fix to emerging issues in a number of key areas, including marketing, sales, customer experience, HR, fulfillment, finance, logistics analytics, and others.

STEPS OF DATA ANALYSIS

a) Identify

Before you get your hands dirty with data, you first need to identify why you need it in the first place. The identification is the stage in which you establish the questions you will need to answer. For example, what is the customer's perception of our brand? Or what type of packaging is more engaging to our potential customers? Once the questions are outlined you are ready for the next step.

b) Collect

As its name suggests, this is the stage where you start collecting the needed data. Here, you define which sources of data you will use and how you will use them. The collection of data can come in different forms such as internal or external sources, surveys, interviews, questionnaires, and focus groups, among others. An important note here is that the way you collect the data will be different in a quantitative and qualitative scenario.

c) Clean

Once you have the necessary data it is time to clean it and leave it ready for analysis. Not all the data you collect will be useful, when collecting big amounts of data in different formats it is very likely that you will find yourself with duplicate or badly formatted data. To avoid this, before you start working with your data you need to make sure to erase any white spaces, duplicate records, or formatting errors. This way you avoid hurting your analysis with bad-quality data.

d) Analyze

With the help of various techniques such as statistical analysis, regressions, neural networks, text analysis, and more, you can start analyzing and manipulating your data to extract relevant conclusions. At this stage, you find trends, correlations, variations, and patterns that can help you answer the questions you first thought of in the identify stage. Various technologies in the market assist researchers and average users with the management of their data. Some of them include business intelligence and visualization software, predictive analytics, and data mining, among others.

e) Interpret

This stage is where the researcher comes up with courses of action based on the findings. For example, here you would understand if your clients prefer packaging that is red or green, plastic or paper, etc. Additionally, at this stage, you can also find some limitations and work on them.

-- THE END --